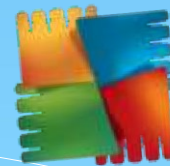


Computer Virus And Anti Virus



*What is a Computer Virus?

A **Computer Virus** is a computer program that can copy itself and infect a computer without permission or knowledge of the user



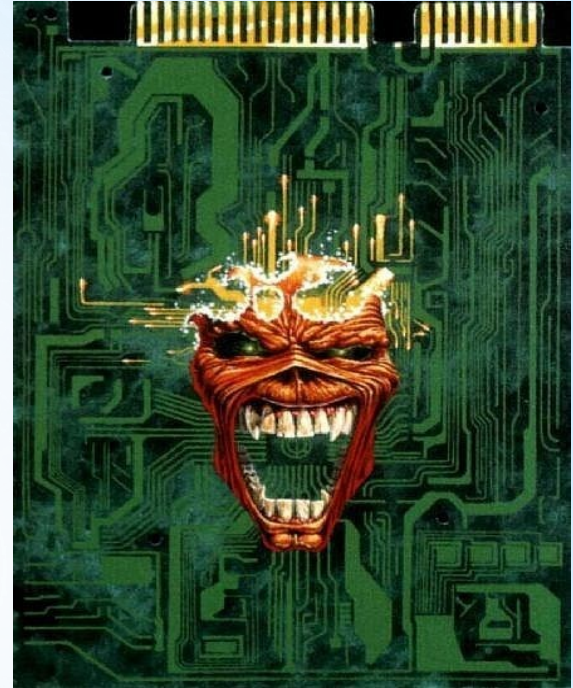
What a computer virus do?

- A virus tries to take control of computer system at the first opportunity available.
- Also it makes copies of it self and try to carry harmful task written in its program.
- This process can happen so quickly that the user is not even aware of the presence of a virus in computer.



* How does viruses spread ?

A **virus** can only spread from one computer to another when its host is taken to the uninfected computer, for instance by a user sending it over a network or the Internet, or by carrying it on a removable medium such as a floppy disk, CD, or USB drive.



History of Virus

Like any other field in computer science viruses have evolved over the years.

- The first computer virus was called 'Creeper'.
- It was invented in the early 1971 by Bob Thomas.
- This program has been designed not to damage the computer, but only to display a messages.
- Since then, millions of viruses have been invented.

Why do People Create Computer Viruses?

Any programmer can make a virus to:

- To take control of a computer and use it for specific tasks.
- To steal sensitive information (credit card numbers, passwords, personal details, data etc.)



Why do People Create Computer Viruses? (continue...)

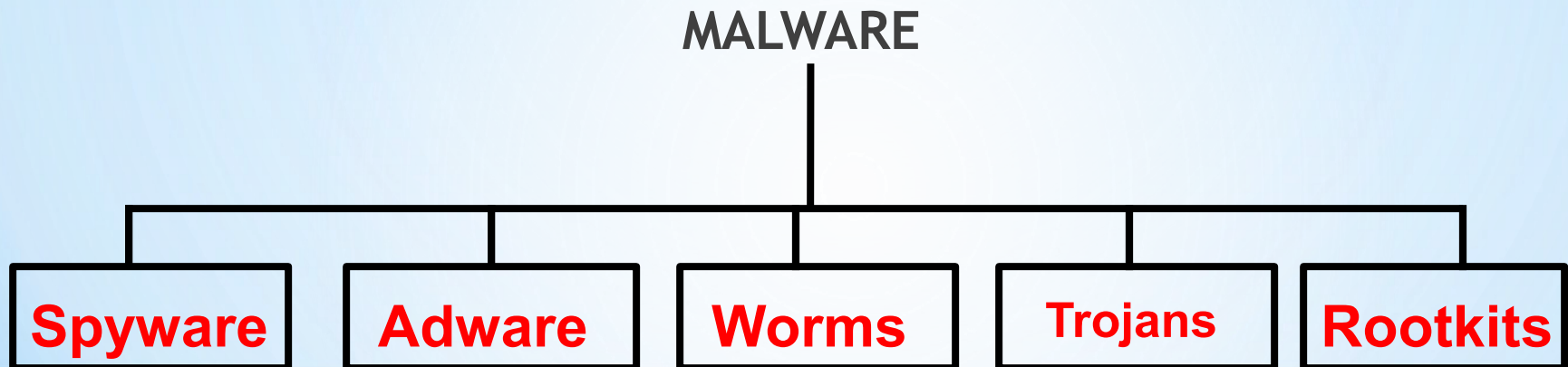
Any programmer can make a virus to:

- To take money.
- To prove ones skill or for revenge purposes.
- To disable a computer or network.



*TYPES OF VIRUSES

On the basis of scale there are five types of viruses under one name:



VIRUSES: WHAT THEY INFECT

1. **System Sector Viruses** -DOS boot sector or the Master Boot Record
2. **File Viruses** -Program (COM and EXE) files
3. **Macro Viruses** -Data files having macro programs
4. **Companion Viruses** -EXE files by installing a same-named COM file
5. **Cluster Viruses** -Changes the DOS directory information so that directory entries point to the virus code instead of the actual program
6. **Source Code Viruses** -Source code, by adding Trojan code to it

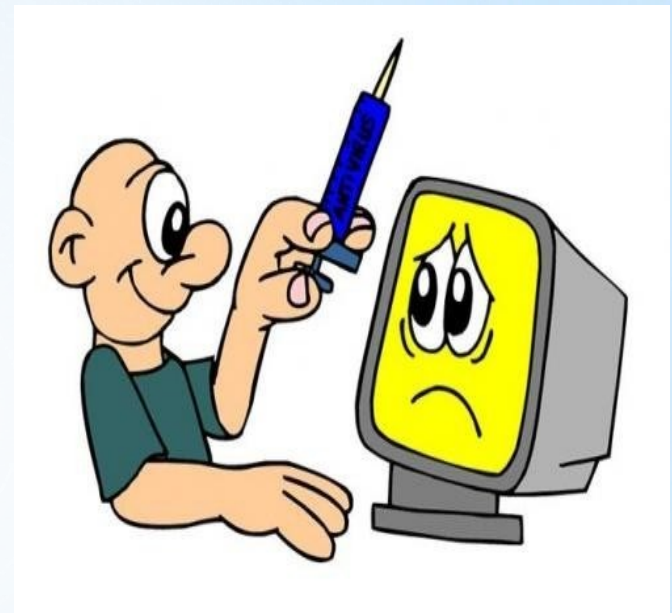
*MALWARE

Malware, from the words malicious and software, is a software designed to damage a computer system without the owner's informed consent. It includes spyware, adware, computer worms and trojan horses.



*SPYWARE

Spyware is computer software that is installed secretly on a personal computer to intercept or take partial control over the user's interaction with the computer, without the user's informed consent.



*COMPUTER WORMS

Computer Worm is a self-replicating computer program. It uses a network to send copies of itself to other nodes (computer terminals on the network) and it may do so without any user intervention. It always cause harm to the network, if only by consuming bandwidth



*TROJAN HORSES

A **Trojan**, is malware that appears to perform a desirable function but in fact performs undisclosed malicious functions



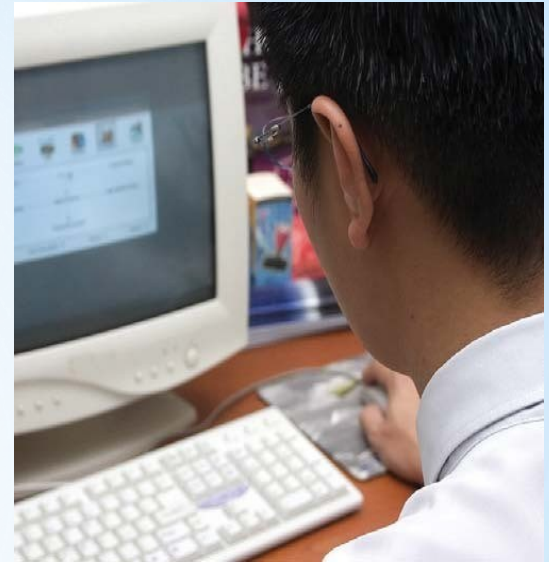
*ROOTKITS

A **Rootkit** is a computer virus which consists of a program designed to take fundamental control of a computer system, without authorization by the system's owners and legitimate managers.



Signs of viruses:

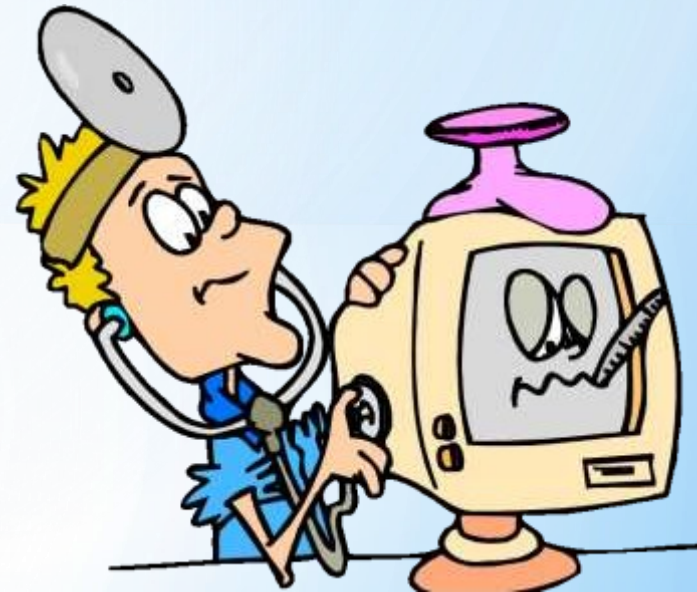
- Operating System may slow down.
- Change the length of executable (*.com/*.exe) files.
- Change to the reassignment of system resources.
- Showing some funny and suspicious behaviour (blinking of the mouse).



How to detect a Virus

Some signs that may indicate that your computer is infected include:

- You see uncommon error messages.
- Applications won't start.
- Your antivirus has disappeared, and firewall is disabled





ANTI - VIRUS

Anti-Virus Software are computer programs that attempt to identify, neutralize or eliminate malicious software. Most modern **antivirus** software is now designed to combat a wide range of threats, including worms, rootkits, Trojans, often described collectively as malware.



How an anti virus works?

- The antivirus software examines each and every file in a computer and examines its content with the virus definitions stored in its virus dictionary.
- A virus dictionary is an inbuilt file belonging to an antivirus software that contains code identified as a virus by the antivirus authors.

* APPROACHES TO DETECT VIRUSES

1. THE SIGNATURE APPROACH:

Just like the police traces patterns in crimes back to a criminal ,software engineers can reverse engineer a virus to find the signature it leaves .this signature is added to the database so that when buster performs a virus scan, each file is scanned for matches with any virus' signature

2.THE SANDBOX APPROACH:

A 'sandbox is an advanced program that emulates an OS. A suspect executable file is run within the confines of a sandbox and then the sandbox is examined to see what changes were made.

3.THE HEURISTIC APPROACH:

Heuristic analysis is a method employed by many computer antivirus programs designed to detect previously unknown computer viruses, as well as new variants of viruses already in the "wild".

How to Protect your System Against Virus?

To Protect your system against Viruses you have to follow these steps :

- Install a good Anti-Virus software on your computer.
- Update Anti-Virus regularly.



How to Protect your System Against Virus? (continue...)

- Be careful while Downloading files or programs from the internet.
- Always scan your floppies, CDs, flash drives before using them.



How to Protect your System Against Virus? (continue...)

- Do not use pirated software.
- Turn on firewall of your computer operating system.



*** Manual Prevention from Viruses**

- 1.Delete your browsing history, temporary internet files, internet cookies before exiting .**
- 2.Always remember to scan any removable hardware using anti-virus software before accessing it.
Removable hardware includes Floppy Disks, CD, USB Drive.**
- 3.Scan your computer every week using anti-virus software like Norton, McAfee for prevention against malware**
- 4.Update your anti-virus for new and latest virus definitions.**
- 5.Create a system restore point before installing any software or freeware.**

Best Antivirus in Market

→ Kaspersky Anti-Virus



→ Norton Antivirus



→ AVG Antivirus



→ NOD32 Antivirus



→ Panda Antivirus



THANK YOU!